

# Partner-Specific Affective Precision in Social Active Inference

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**Abstract.** In multi-agent social settings, behavior and model reliability vary significantly across relationships. Beyond accurately inferring what other agents will do, an agent must also calibrate how confidently to act on those inferences—and do so independently for each relationship. An agent may carry a well-validated model of one partner, a fragile model of another, and an actively revising model of a third; collapsing these into a single confidence estimate loses information that matters for action. Affect has traditionally been described in terms of valence and arousal, but it can also serve a functional role: regulating how strongly beliefs are expressed in action rather than shaping their content. Using the active inference framework, we model affect as a metacognitive estimate of confidence in one’s current partner model, and formalize this by extending affective precision to be relationship-specific. Each partner’s behavioral evidence updates its own confidence estimate, which modulates how strongly the agent commits to its inferred partner state during action selection. Simulations in a multi-partner trust game show that partner-local affective precision tracks predictive reliability rather than realized value: a reliably cooperative and a reliably defecting partner can both support high precision, because what they share is predictability, not positive value. In open decision regimes, behavioral effects arise through action sharpening rather than improved partner-state inference. Confidence supports decisive action when the partner model remains valid, but can impair adaptation when a partner changes abruptly, indicating that affective precision is a calibration mechanism rather than a guaranteed payoff-improvement mechanism. Varying precision gain and prior model fitness reveals a family of trust-calibration profiles, providing a computational basis for individual differences in social trust. Together, these results describe affective precision as a relationship-specific metacognitive signal that separates social prediction from social commitment.

**Keywords:** Active inference · Affective precision · Social metacognition · Multi-agent systems · Trust · Individual differences

## 1 Introduction

Social agents rarely act on a single, stable model of the social world. They maintain different expectations for different partners: one collaborator may be

familiar and reliable, another newly encountered and uncertain, and another predictable but adversarial. Repeated social exchange depends on learning such partner-specific regularities, including reciprocity, reputation, moral character, and social value [15, 3, 7]. In such settings, the question is not only what an agent believes about each partner, but how confidently those beliefs should guide policy inference.

This makes social decision-making useful to separate into three computational problems. The first is inferring what a partner is like—whether they tend to cooperate, defect, or behave unpredictably. The second is modeling what they know or intend—the theory-of-mind question of how another agent represents the world and plans action [11, 12, 18]. The third is estimating whether one’s current partner model is reliable enough to guide confident action. Accurate type inference and sophisticated mentalizing do not by themselves solve this third problem. Knowing that someone is unreliable is not the same as knowing how cautiously to treat them; knowing that someone is predictable is not the same as knowing how strongly to commit to that prediction. An agent that infers accurately but commits poorly wastes its social knowledge; one that commits confidently on fragile models exposes itself to exploitation. Many formal models specify what agents infer about others while treating confidence in those inferences as a fixed or global quantity.

We model social affect as a metacognitive estimate of whether the focal agent’s current partner model warrants confident action, rather than as a direct readout of another agent’s future behavior or a record of reward and punishment. In psychology and neuroscience, affect is often described in terms of valence and arousal—the felt pull toward or away from situations and social commitments [2]. Functionally, however, it can regulate how strongly beliefs are expressed in action: affect assigns action-relevant weight to what an agent already infers, shaping the degree of commitment rather than the content of inference.

The active inference framework [9, 10, 17] provides a formal route from this observation to a computational mechanism. In active inference, policy inference is precision-weighted: the same beliefs can produce decisive or tentative behavior depending on  $\gamma$ , the parameter controlling how sharply an agent commits to its inferred policy distribution. Hesp et al. [14] made affect endogenous by modeling it as inference over a higher-order belief  $\beta$  representing subjective model fitness—an affective charge signal  $\phi$  drives updates to  $\beta$ , rising when the model performs better than a prior baseline and falling when it performs worse. On this view, affect is metacognitive rather than a direct representation of another agent’s intentions: it tracks model fitness rather than partner intentions or reward history alone. Recent work on theory of mind in active inference [18, 19] addresses how agents model what others know or intend through recursive belief structures and factorized generative models—solving problem two well, but assuming rather than estimating action confidence. Hierarchical volatility models [16, 3] and related applications to social learning [8] address adaptive belief updating under uncertainty by estimating when prediction errors should drive larger or smaller revisions to belief.

What none of these approaches address is whether confidence in a social model should itself be relationship-specific. In multi-partner settings, model fitness can differ sharply across partners—a person can carry a well-validated model of a longtime collaborator, a fragile model of a new acquaintance, and an actively revising model of someone who has recently behaved unexpectedly. Collapsing these into a single shared precision budget loses information that matters for action. We therefore extend affective precision to be relationship-specific: each partner receives its own  $\beta$  estimate, updated only from that partner’s behavioral evidence, which modulates policy precision when inferring policies toward that partner.

We evaluate this mechanism in a repeated multi-partner trust game across binary and graded decision regimes, partner-choice conditions, abrupt betrayal, and profile-style parameter variation. The simulations evaluate three dissociations: that partner-local affective precision tracks predictive reliability more than realized value; that behavioral effects arise through action sharpening rather than improved partner-state inference; and that precision gain together with prior model fitness shapes temporally structured profiles of social trust revision.

The remainder of the paper is organized as follows. Section 2 specifies the multi-partner trust game, per-partner generative models, and the partner-local affective precision mechanism. Section 3 reports simulation results. Section 4 discusses implications, limitations, and directions for future work.

## 2 Methods

A focal active-inference agent plays a repeated trust game with four partners whose behavioral types and stances are hidden and must be inferred from behavioral evidence. The agent is implemented using the `pymdp` library for discrete active inference [13, 6], which provides standard components for belief updating and policy selection under the active inference framework [9, 10, 17]. Partner-local affective precision is layered on top of per-partner generative models, allowing evidence and action confidence to remain relationship-specific. Full POMDP specifications, affective precision update rules, and simulation protocols are provided in Appendix A, Appendix B, and Appendix C, respectively.

### 2.1 Trust game

The trust game [4, 15] provides a natural testbed because partner types are hidden and must be inferred, creating a genuine gap between social knowledge and actionable confidence. Each partner has a latent *type*—cooperator, reciprocator, exploiter, or random—and a latent *stance* toward the agent—trusting, neutral, or hostile—neither directly observable; both must be inferred from behavioral evidence. Episodes run for up to 200 rounds.

In the binary game both players choose to cooperate or defect, with focal agent payoffs as shown in Table 1. The structure poses a social inference problem: defection is locally tempting, but cooperative partners are worth identifying and

maintaining over repeated interaction. In the graded regime, the binary choice is replaced by a discretized investment level. In the partner-choice regime, the agent selects which partner to engage before choosing an action each round, making approach and avoidance explicit.

Table 1: Focal agent payoffs in the binary trust game.

|                  | Partner cooperates | Partner defects |
|------------------|--------------------|-----------------|
| Agent cooperates | 3                  | -1              |
| Agent defects    | 5                  | 1               |

Reported experiments use one focal active-inference agent interacting with four environment-side partners. Partners sample actions from type-by-stance cooperation probabilities and undergo action-dependent stance transitions, but do not perform variational inference or maintain affective precision estimates. This isolates the focal agent’s precision mechanism before extending to reciprocal active-inference partners.

## 2.2 Per-partner generative models

Rather than maintaining a single pooled social model, the focal agent maintains a separate discrete generative model  $\mathcal{M}_k$  for each partner  $k$ . This factorization keeps evidence about each partner independent and allows action confidence to vary across relationships. Each model encodes observation likelihoods, transition dynamics, preferences, priors, and policy priors using the standard  $A/B/C/D/E$  structure of discrete active inference [6]. Full matrix specifications are provided in Appendix A.

Each partner model contains three latent factors: partner type, partner stance, and a deterministic own-action bookkeeping factor that makes the agent’s executed action available to the payoff likelihood. Observations are partner action and focal payoff. The partner-action likelihood is defined by type- and stance-conditioned cooperation probabilities, operationalizing predictive reliability: both a reliably cooperative and a reliably defecting partner can be predictable, even though their payoff consequences differ. Payoff enters as a second observation modality, with partner action marginalized through the cooperation likelihood, so preferences over outcomes remain part of the generative model rather than an external reward cache. Stance transitions are action-dependent—cooperation tends to build trust, defection damages it, and recovery from hostility is slower than collapse.

## 2.3 Partner-local affective precision

Hesp et al. [14] treat policy precision as an inferred quantity, modeling affect as active inference over a higher-order belief  $\beta$  representing subjective model

fitness. We extend this to the per-partner case by assigning each partner its own  $\beta_k$  estimate, updated from that partner’s behavioral evidence alone.

After interacting with partner  $k$ , the agent computes surprisal from the predicted probability of the observed partner response:

$$\epsilon_k = -\log P(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \mid q(s_k^{\text{type}}, s_k^{\text{stance}})). \quad (1)$$

This is computed from partner action rather than payoff because action is the most direct test of type-by-stance model fit: a reliably defecting partner produces no action surprise even though the payoff is poor. The neutral fitness baseline is set at  $\sigma_0^2 = (-\log 0.5)^2$ , so a fifty-fifty binary prediction carries zero affective charge. Affective charge is then:

$$\phi_k = \alpha(\sigma_0^2 - \epsilon_k^2), \quad (2)$$

where  $\alpha$  is a gain parameter. Positive  $\phi_k$  means partner  $k$ ’s model predicted better than baseline; negative  $\phi_k$  means it predicted worse. A categorical posterior  $q(\beta_k)$  is updated each round by combining a slowly evolving persistence prior with a charge-based pseudo-likelihood; full update rules are given in Appendix B. The agent then sets policy precision from the posterior mean:

$$\gamma_k = \gamma_{\text{base}} / \mathbb{E}_{q(\beta_k)}[\beta_k], \quad (3)$$

where  $\gamma_{\text{base}}$  is the baseline policy precision applied in the absence of affective modulation. Here  $\beta_k$  represents inverse precision: higher  $\beta_k$  means the model has been less reliable, producing a broader, less committed policy distribution. Accordingly, positive model fitness is expressed as lower expected  $\beta_k$  and therefore higher policy precision  $\gamma_k$ ; when we refer below to precision or confidence tracking reliability, we mean this  $\beta_k$ -derived policy-precision signal rather than raw  $\beta_k$  increasing with reliability. The mechanism thus translates accumulated evidence about predictive reliability into action confidence independently for each social relationship. The  $\beta_k$  posterior is maintained as an auxiliary precision tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent’s ability to form prior expectations over its own future affective states; the implications of this design choice are discussed in Section 4.

## 2.4 Cross-partner policy selection

In partner-choice regimes, policies specify both which partner to engage and how to act toward that partner. A naive application of partner-local precision creates an artifact: low precision toward a uniformly poor partner branch scales its logits toward zero, making it spuriously attractive in cross-partner comparison.

We address this by applying precision to centered branch logits:

$$\tilde{u}_{k,\pi} = \bar{u}_k + \gamma_k (u_{k,\pi} - \bar{u}_k), \quad (4)$$

where  $u_{k,\pi}$  is the pre-precision policy logit for policy  $\pi$  toward partner  $k$  and  $\bar{u}_k$  is that branch’s mean logit. Centering preserves the mean score of each partner branch while allowing  $\gamma_k$  to sharpen or flatten the within-partner policy distribution. The agent selects from the combined set of centered logits across all partner branches. Full details are provided in Appendix B.

## 2.5 Simulation setup

Simulations compare four affect-runtime variants: full partner-local precision, a shared- $\beta$  ablation that pools evidence across partners while keeping per-partner POMDP beliefs intact, a tracked-only ablation that updates  $q(\beta_k)$  but fixes  $\gamma_k = \gamma_{\text{base}}$ , and a no-affect control that disables the  $\beta$  tracker entirely. Conditions span binary and graded decision regimes, partner-choice settings, abrupt betrayal, and profile-style parameter variation across precision gain  $\alpha$  and prior model fitness. Central comparisons use 30 seeds per condition; profile-scale experiments use 20 (Appendix C). Full condition definitions, episode lengths, and metrics are provided in Appendix C. Simulation code, analysis scripts, and experiment configurations will be released in a public repository upon acceptance.

## 3 Results

The experiments are organized around three dissociations. The first tests whether  $\beta_k$ -derived policy precision tracks predictive reliability independently of realized value, and whether pooling evidence across partners destroys this relationship-specific signal. The second tests whether precision affects behavior through action commitment rather than improved partner-state inference. The third examines how the mechanism behaves under social change, and how precision gain and prior model fitness shape the temporal structure of trust revision. Each dissociation is evaluated by varying one element of the design while holding others fixed. We therefore treat payoff as a downstream diagnostic of how confidence calibration interacts with task structure, not as the primary criterion for whether the mechanism is present.

### 3.1 Precision tracks predictability, not partner value

The affective charge formula is grounded in action surprisal, not payoff. This means the inverse-precision posterior  $q(\beta_k)$  tracks behavioral consistency: reliable predictions shift mass toward lower  $\beta_k$  and higher  $\gamma_k$ ; unreliable predictions shift it the other way. The consequence is non-obvious. Once the agent has inferred a partner’s type and stance, a reliably defecting partner can generate low surprisal despite poor payoff, while an erratic cooperator can generate high surprisal despite favorable outcomes. By construction, the precision signal tracks the quality of the agent’s predictive model—not the desirability of the partner’s behavior. The 30-seed model-fitness analysis tests whether this holds across the simulated social portfolio.

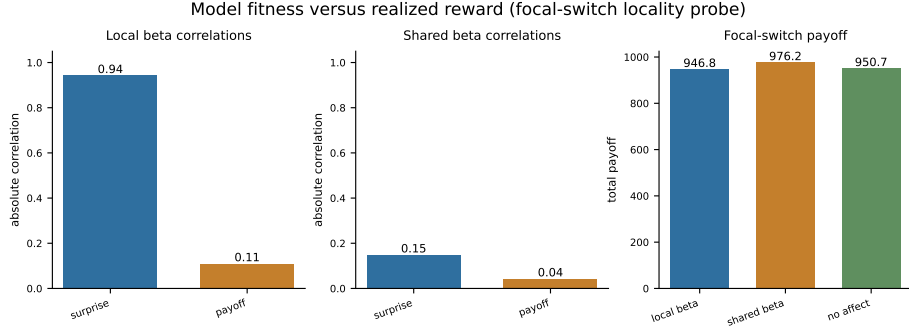


Fig. 1: Partner-local precision tracks action surprisal more strongly than realized payoff. The panel contrasts active-encounter partial associations between mean partner precision and surprise versus payoff for partner-local  $\beta_k$  and a shared  $\beta$  tracker, plus aggregate payoff. Partner-local tracking preserves the surprise-over-reward dissociation; pooling evidence across partners attenuates relational specificity.

Taken raw, partner-local precision associates more strongly with action surprisal than realized payoff (0.949 versus 0.748). But in partner-choice episodes, payoff, surprisal, and exposure are correlated—partners engaged more often generate more evidence on all three dimensions simultaneously. Controlling for encounter imbalance sharpens the story: the precision–surprisal association holds at 0.882, while the precision–payoff association falls to 0.130. The mechanism is doing what it was designed to do.

Pooling evidence across partners attenuates this. A shared  $\beta$  tracker preserves the coarse ordering but compresses both associations toward weaker values (0.380 versus 0.049), confirming that the partner-specific signal depends on locality, not just the update rule. The aggregate payoff readout adds a sharper point: cumulative reward ranks inversely to the mechanism’s presence—no-affect achieves the highest payoff (542.1), followed by shared (517.6) and partner-local (483.5). Affective precision is not optimizing reward. It is calibrating action confidence. Figure 1 shows this dissociation directly.

### 3.2 Shared precision tests relational specificity

The reliability-over-value dissociation in Section 3.1 depends on evidence remaining tied to the partner who generated it—but whether that is because of locality specifically, or simply because any accumulating precision signal would behave this way, needs ruling out. The shared- $\beta$  ablation tests this directly. It applies the same update rule to pooled evidence, keeping per-partner POMDP beliefs intact. If the signal degrades under pooling, the update rule alone is not sufficient; the relational structure of the evidence is doing the work.

It degrades. Under shared  $\beta$ , the surprise association falls from 0.882 to 0.380 and the payoff association from 0.130 to 0.049—both compressed, both weakened. Locality is load-bearing. Pooling evidence blurs model-fitness estimates across relationships, converting a partner-specific confidence signal into a property of the social environment as a whole. Whether that blurring costs the agent behaviorally depends on how heterogeneous the partner portfolio is—a question the mixed-volatility analyses return to directly.

### 3.3 Behavioral effects arise through action sharpening

Does partner-local affective precision change behavior by sharpening how existing beliefs are deployed in action—or does it improve the beliefs themselves? These explanations are separable. A tracked-only ablation updates  $q(\beta_k)$  normally but holds  $\gamma_k$  fixed at  $\gamma_{\text{base}}$ , leaving the precision estimate intact while cutting it off from action selection. If the behavioral effect survives, it comes from improved partner-state inference. If it disappears, the effect lives in the  $\beta_k \rightarrow \gamma_k$  deployment step.

It disappears. In the open graded regime, tracked-only matches the no-affect baseline on both entropy (8.79 versus 8.79) and payoff (1864.2 versus 1864.2). Full precision modulation is the only variant that lowers entropy (8.59), while posting a slightly lower cumulative payoff (1851.3). The entropy gap is modest, but its locus is clean: the same beliefs, deployed more sharply, produce more committed behavior—and that commitment does not automatically improve reward. This rules out the interpretation that the  $\beta_k$  tracker improves partner-state inference as a side effect. The beliefs are unchanged. What changes is the degree of commitment they produce. Figure 2 shows the three-way contrast.

That affective precision lowers entropy without a payoff gain is not a failure of the mechanism. It is consistent with viewing affective precision as a calibration layer—one that regulates how confidently the agent acts on what it knows, not one that guarantees better outcomes from acting more confidently. Whether sharper deployment translates to payoff advantage depends on what the agent is being confident about, and toward whom—a question the partner-choice condition addresses directly.

### 3.4 Partner selection also sharpens under precision modulation

The previous section showed that action-sharpening occurs within a fixed partner interaction. Does the same effect shape *which* partner the agent approaches? In partner-choice regimes, the agent must commit to a partner before committing to an action—making partner selection the upstream decision that action sharpening then operates on.

Policy entropy drops under precision modulation here too (3.99 versus 4.83), confirming that the  $\beta_k \rightarrow \gamma_k$  pathway reaches partner selection, not just action selection. The allocation shift is small but directionally consistent with the mechanism: the partner-local agent pulled slightly toward the cooperator (36.6% versus 34.8%) and away from the exploiter (13.8% versus 16.2%)—the



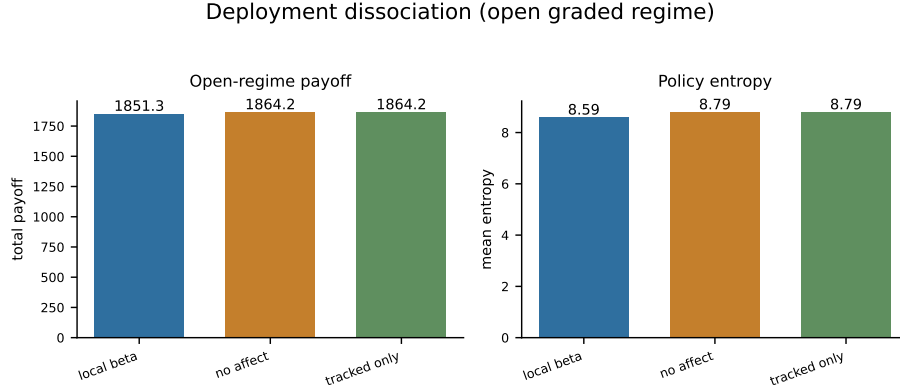


Fig. 2: Open graded regime: partner-local affective precision lowers policy entropy relative to no-affect and tracked-only controls without a corresponding payoff advantage. Tracked-only updates  $\beta_k$  but fixes  $\gamma_k$ ; its deployment readouts match the no-affect baseline, isolating the entropy effect to the  $\beta_k \rightarrow \gamma_k$  action-confidence pathway.

partners the mechanism would assign higher and lower confidence to, respectively. Aggregate payoff does not separate at this replication scale (393.6 versus 393.2), extending the dissociation from Section 3.1: the mechanism is not redistributing engagement toward higher-value partners—it is sharpening commitment toward more predictable ones.

Whether this redistribution produces payoff consequences depends on whether partner behavior stays stable. A one-round reallocation toward a predictable cooperator is worth little if that partner changes behavior mid-episode. The abrupt-betrayal condition tests exactly this.

### 3.5 Abrupt betrayal tests confidence under change

Accumulated confidence is useful when it matches the social environment. When a partner changes, it can become a liability—the agent enters the post-switch period with evidence that was valid under a relationship that no longer exists. The abrupt-betrayal condition tests this directly: a previously cooperative partner switches stance at round 31, and the agent must detect the change and reallocate while carrying confidence built under the old model.

The clearest result is in entropy. Under precision modulation, mean policy entropy is lower than the no-affect control (8.36 versus 8.74), with a bootstrap interval that does not cross zero (−0.62 to −0.14). The action-confidence pathway is active and its effect is reliable at this scale. Joint partner-type accuracy is also higher under precision modulation (0.372 versus 0.266; bootstrap interval 0.034 to 0.185)—the agent is tracking engaged partners more accurately while managing

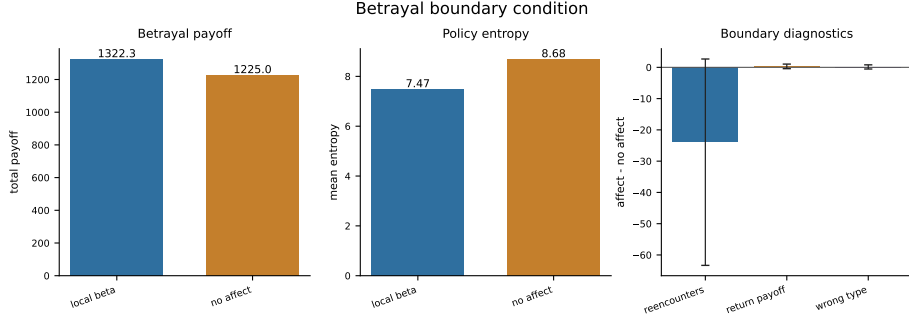


Fig. 3: Abrupt betrayal in the partner-choice regime (30 seeds; P0 stance switch at round 31). Partner-local affective precision lowers policy entropy relative to the no-affect control; joint partner-type accuracy is higher under precision modulation. The payoff difference is positive but its bootstrap interval crosses zero—consistent with a calibration mechanism whose deployment effects are reliable while payoff consequences depend on reallocation dynamics.

the post-switch reallocation, consistent with engagement being redirected rather than simply disrupted.

The payoff difference is positive but uncertain (1185.9 versus 1172.1; bootstrap interval  $-25.2$  to  $53.2$ ). This is not a power failure—it is the correct result for a calibration mechanism. Whether accumulated confidence helps or hurts payoff after a switch depends on how quickly engagement can be redirected and toward which partners. At this replication scale, that variance dominates. The mechanism reshapes deployment reliably; whether that deployment pays off depends on the portfolio structure it is reallocating within. Figure 3 summarizes the post-switch readouts.

What this reveals is that the *speed* of confidence revision is not a fixed property of the mechanism—it is controlled by precision gain and prior model fitness. Agents that revise quickly may reallocate faster but accumulate false positives toward stable partners; agents that revise slowly may sustain commitment errors longer after a switch. The profile analyses vary these parameters directly to examine how the confidence-revision process differs across agents.

### 3.6 Precision dynamics as computational profiles of social trust calibration

Four experiments vary the parameters that govern how quickly and how strongly confidence revises: a gain sweep across eight  $\alpha$  values, a factorial crossing of prior belief against  $\alpha$ , a forgiveness paradigm where a betraying partner reverts, and a mixed-volatility environment where partners differ in stability. Together they ask whether social trust calibration varies along a single axis—and show that it does not.

*Precision-gain axis.* Sweeping  $\alpha$  across  $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$  confirms that gain controls the amplitude of  $\beta_k$  dynamics: range increases monotonically with  $\alpha$  in the betrayal regime, from 0.147 at  $\alpha = 0.05$  to 1.218 at  $\alpha = 8.0$ . But payoff does not follow. Across both betrayal and open graded regimes, cumulative reward shows no monotonic relationship with gain (range 1914.6–1986.2 and 1906.3–1987.7 respectively). Faster confidence revision is not better confidence revision. The behavioral consequence of gain depends on what the agent is revising toward—a question that requires varying prior structure as well. Extended alpha-sweep diagnostics are in Appendix D.

*Gain-prior profiles.* Crossing prior belief (naive versus cautious) with  $\alpha$  (low versus high) defines four profiles tested in the betrayal environment (20 seeds). Each isolates a different failure mode. The naive-stubborn agent (naive prior, low  $\alpha$ ) shows the most diffuse partner selection (Gini 0.304) and the slowest confidence dynamics ( $\beta_k$  range 0.36)—it commits weakly everywhere and revises slowly. The avoidant-rigid agent (cautious prior, low  $\alpha$ ) posts the lowest payoff (1889) despite identical gain: the cautious prior suppresses confidence even toward reliably cooperative partners, so the agent never learns to exploit predictability. The anxious-reactive agent (naive prior, high  $\alpha$ ) accumulates confidence quickly ( $\beta_k$  range 1.08) but pays for it—it shows the highest post-betrayal commitment errors toward the switched partner (P0 selection 0.319; high investment rate 0.121), because the same speed that builds confidence fast also sustains it too long after a partner changes. The default agent achieves the highest payoff (1991), not because any single parameter is optimal, but because its gain and prior sit in a region that balances these competing costs. Quadrant figures and supporting diagnostics are in Appendix D.

*Forgiveness and trust repair.* Forgiveness separates two things that might seem equivalent: willingness to reengage a partner who has reverted, and restoration of the confidence that partner’s model once supported. They come apart. After the betrayed partner reverts to cooperative behavior at round 121, no-affect and cautious-low- $\alpha$  agents reengage most readily (selection rates 0.593 and 0.630)—consistent with weaker affective commitment and muted  $\beta_k$  dynamics (cautious-low  $\beta_k$  range 0.29 versus default 1.01). High-gain profiles show larger  $\beta_k$  swings but not faster payoff recovery; across all variants, payoff recovery sits near baseline (0.996–1.033). The agent can return to a partner before it trusts them again. These are different things, and the mechanism tracks both independently.

*Mixed-volatility discrimination.* The mixed-volatility environment poses a simultaneous test: maintain stable engagement with a stationary cooperator (P0) while updating away from an abrupt-switch partner (P2) and a gradual-drift partner (P3), all within the same episode. Discriminating between stable and shifting partners requires the confidence signal to move in opposite directions for different relationships at the same time.

Table 2: Mixed-volatility metrics by precision profile (20 seeds).

| Variant        | Payoff | Discrimination | P0 false positive | $\beta$ range |
|----------------|--------|----------------|-------------------|---------------|
| default        | 1976   | 0.121          | 0.422             | 0.933         |
| high- $\alpha$ | 2054   | -0.089         | 0.484             | 1.049         |
| low- $\alpha$  | 1927   | 0.213          | 0.359             | 0.182         |
| no-affect      | 1934   | -              | 0.273             | -             |

Higher gain produces higher payoff in this environment (high- $\alpha$ : 2054 versus default: 1976 versus low- $\alpha$ : 1927)—but also more false positives. The rate of unwarranted confidence reduction toward the stable P0 increases with  $\alpha$  (0.484 versus 0.422 versus 0.359), and the discrimination index does not improve monotonically: low- $\alpha$  shows the strongest discrimination (0.213) while high- $\alpha$  goes negative (-0.089). Gain amplifies reactivity in both directions. In a mixed-volatility portfolio, higher gain raised mean payoff while worsening discrimination—payoff and partner-specific calibration therefore dissociated rather than improving together. Figure 4 shows the  $\beta_k$  trajectories that underlie this pattern.

Across these four experiments, gain and prior initial conditions jointly determine how an agent builds, maintains, and revises social confidence—and the dimensions they control trade off rather than co-vary. High gain accelerates both useful updating and false positives. Low gain produces stability at the cost of exploitation speed. Cautious priors protect against premature trust while suppressing confidence even where it is warranted. No single profile dominates across environments. This is why the paper treats any single profile as a computational description of a trust-calibration strategy rather than a prescription for optimal behavior. Extended profile tables, alpha-sweep diagnostics, and forgiveness/trust-repair analyses are provided in Appendix D.

## 4 Discussion

The results establish three dissociations: precision tracks predictive reliability over realized value, behavioral effects arise through deployment rather than improved inference, and partner-local confidence carries information that shared confidence loses. Each dissociation isolates a different aspect of the same underlying claim—that social prediction and social commitment are separable problems that an agent can solve independently for each relationship. What the simulations add beyond this design intention is the shape of the tradeoffs. Calibration is not a free lunch. Sharper deployment does not reliably improve payoff. Accumulated confidence can become a liability under change. The parameters that govern how quickly confidence revises do not co-vary along a single beneficial axis. Payoff appears here as a diagnostic of how calibration interacts with task structure, not as the primary criterion for whether the mechanism is present—because a calibration mechanism should be evaluated first on whether it calibrates, and second on whether good calibration helps in a given environment.

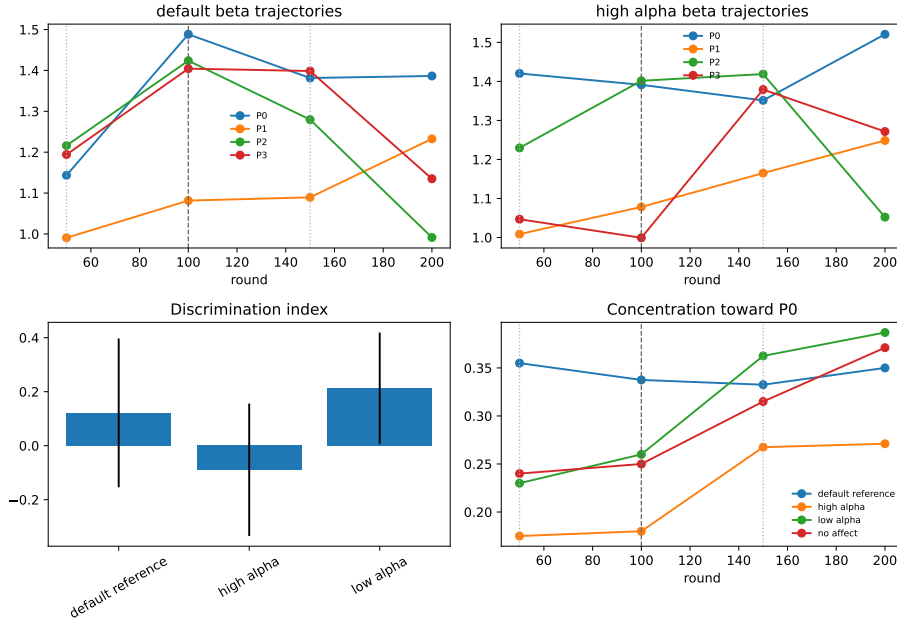


Fig. 4: Mixed-volatility partner-choice environment (20 seeds). Four partners differ in volatility: stationary cooperator (P0), stationary exploiter (P1), abrupt switch (P2), and gradual drift (P3). Higher precision-gain  $\alpha$  can raise payoff while worsening discrimination and increasing false-positive confidence reduction toward P0.

The reliability-over-value dissociation follows directly from the update rule, and it has a non-obvious implication worth stating plainly. A reliably defecting partner and a reliably cooperative partner are computationally equivalent from the perspective of affective precision—both are predictable, so both support high action confidence. What differs is what the agent does with that confidence, not whether it has it. This is the sense in which affect here is metacognitive rather than evaluative: it reflects how well the agent knows what is happening, not how good what is happening is. This separates the mechanism from reward tracking, from valence-based accounts of emotion [2], and from policy selection mechanisms that weight actions by their expected payoff directly. The agent can be confident about a bad partner. It can be uncertain about a good one. Confidence and desirability are orthogonal signals, and the mechanism keeps them that way by construction.

The tracked-only ablation shows that the precision estimate only matters insofar as it reaches action selection. When  $\beta_k$  was updated but  $\gamma_k$  was held fixed, behavior was indistinguishable from the no-affect baseline. This rules out the possibility that the precision tracker improves partner-state inference as a

side effect—the beliefs themselves are unchanged. What changes is the degree of commitment: the same inferred partner state produces sharper action under higher confidence. Affect here operates downstream of inference, on the mapping from beliefs to action rather than on the beliefs themselves.

This positions the mechanism precisely relative to existing accounts. Hesp et al. [14] made affect endogenous to active inference by modeling it as inference over a higher-order belief representing global model fitness—the present extension makes that estimate relationship-specific, so that confidence becomes a property of individual social bonds rather than a global affective state. Theory-of-mind accounts in active inference [18] address what another agent knows or intends; partner-local affective precision addresses how confidently the focal agent should act on its own current model of that partner. These problems are complementary rather than competing—a richer agent would use theory-of-mind inference to improve the *content* of its partner model while using partner-local affective precision to regulate the *confidence* with which that model is deployed. Volatility models such as the HGF [16] regulate how quickly beliefs update when the environment changes; the present mechanism regulates how strongly current beliefs are expressed in action. The two can be connected—a change-detection layer that discounts stale confidence when prediction-error structure shifts would address the main limitation exposed by the betrayal and mixed-volatility conditions—but they are not the same operation.

The betrayal and mixed-volatility experiments reveal where the mechanism’s assumptions break down. Partner-local affective precision is well-suited to stable social environments, where accumulated confidence correctly reflects a relationship that still holds. Under abrupt change, the same accumulated confidence misfires: the agent enters the post-switch period with evidence that was valid under a relationship that no longer exists. The most consistent betrayal effect is deployment change rather than payoff improvement, and payoff consequences depend on how quickly engagement can be reallocated—a quantity the mechanism does not directly control. The mixed-volatility results sharpen this: higher gain can increase payoff while worsening discrimination and generating false-positive confidence reductions toward stable partners. Reactivity cuts in both directions. Partner-local affective precision should be understood as a first calibration layer, not a complete social-control architecture.

The natural architectural complement is therefore a volatility or change-detection layer that monitors shifts in prediction-error structure and discounts stale affective confidence when a previously reliable partner begins behaving unexpectedly. This is also where the mechanism differs from standard volatility models: volatility models regulate how quickly beliefs update when the environment changes, whereas the present mechanism regulates how strongly current beliefs are expressed in action. A change-detection layer could connect the two—monitoring prediction-error structure and discounting stale affective confidence when change is detected. This would preserve the core benefit of the current mechanism—relationship-specific action confidence—while reducing the post-change liability of overcommitting to outdated partner models. In

performance-oriented multi-agent systems, this would turn affective precision from a deployment modulator into part of a broader adaptive control loop: predict partner behavior, estimate confidence in that prediction, detect when confidence has become stale, and reallocate exploration or commitment accordingly.

The profile experiments use parameter combinations as computational analogues for empirically studied variation in social trust calibration [3, 1, 5], not as fits to human behavioral data. Precision gain and prior model fitness jointly determine update speed, confidence amplitude, sensitivity to change, reengagement after repair, and vulnerability to false positives—and these dimensions trade off rather than co-vary. High gain can increase payoff while worsening discrimination; low gain produces stable engagement but slow recovery from betrayal; cautious priors suppress confidence even toward reliable partners. The profiles generate concrete behavioral predictions: an anxious-reactive agent (naive prior, high gain) should show rapid confidence accumulation followed by high post-betrayal commitment errors, while an avoidant-rigid agent (cautious prior, low gain) should show muted confidence dynamics and poor payoff even with cooperative partners. Whether these profiles correspond to stable individual differences in human social trust—connecting to work on social learning [3], psychosis [1], and attachment [5]—remains an empirical question that will require fitting the model to behavioral time series. For artificial agents, the same profiles suggest engineering tradeoffs: high-gain agents may react quickly but risk false positives and post-betrayal overcommitment; low-gain agents may remain stable but fail to exploit reliable partners quickly; cautious priors may protect against premature trust while suppressing useful confidence.

Several architectural extensions would turn the present calibration mechanism into a more complete and performance-oriented social agent. Most foundationally,  $\beta_k$  is currently maintained as an auxiliary tracker outside the generative model; embedding it as a hidden state would allow the agent to form prior expectations over its own future action confidence and choose policies partly by considering how that deployment confidence may evolve [14]. A volatility or change-detection layer could discount accumulated confidence when prediction-error structure shifts, addressing the main limitation exposed by the betrayal and mixed-volatility conditions. Partner-choice policies could be made explicitly exploratory after detected change, enabling the agent to sample alternatives rather than simply sharpening or flattening action toward the current partner. Most consequentially for social complexity, environment-side partners could be replaced with reciprocal active-inference agents that maintain their own beliefs, policies, and precision estimates—allowing partner-local affective precision to interact with theory-of-mind inference in a fully reciprocal multi-agent setting.

Two empirical extensions follow from the profile results. The model produces behavioral quantities measurable from trust-game time series—exploitation sensitivity, betrayal recovery time, reengagement latency after repair, false-positive confidence reduction, and partner reallocation under mixed volatility. Fitting  $\alpha$  and prior model-fitness parameters to individual participants would test whether the computational profiles correspond to stable individual differences in social

trust calibration. The dissociations that are most robust at the current simulation scale—predictability over value, deployment over inference, and partner-local over shared confidence—are the natural targets for empirical validation first. Only after these dissociations are validated should future work ask whether richer versions of the architecture produce robust performance gains in more realistic social environments. The mechanism earns the right to be more ambitious by first being right about what it already claims.

## 5 Conclusion

Social affect has typically been described in terms of valence and arousal—the felt pull toward or away from situations and commitments. The results here support a different functional reading: affect as a running metacognitive estimate of how confidently the agent should act on its current model of each partner. Implemented as partner-local affective precision, this signal tracks predictive reliability rather than realized value, reshapes action deployment without altering the beliefs that deployment draws on, and produces temporally structured profiles of trust revision whose consequences for payoff depend on portfolio structure and reallocation dynamics.

The three dissociations—predictability over value, deployment over inference, partner-local over shared confidence—each isolate a different aspect of the same separation: knowing what a partner will do, and committing to that knowledge in action, are problems an agent can solve independently for each relationship. Precision gain and prior model fitness define a family of computational profiles along which that separation plays out differently—varying how quickly confidence builds, how far it swings under change, how readily it recovers after repair. No single profile dominates. That is the point: social trust calibration is not a dial to be turned up, but a multidimensional process whose tradeoffs depend on the social environment an agent inhabits.

The mechanism described here is a first layer. Volatility detection, adaptive confidence discounting, and reciprocal partner models would each extend its reach—but the more fundamental extension is empirical. The profiles generate predictions about behavioral time series that can be tested, and the dissociations established in simulation are the right targets to validate first. Whether partner-local affective precision, as a computational account of social metacognition, maps onto the structure of human social trust is the question the present work opens.

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## Appendix

### A Full POMDP Specification

This appendix specifies the POMDP components used by the focal agent.

#### A.1 Hidden state factors and observations

As described in Section 2.2, the focal agent’s generative model for each partner  $k$  maintains three latent factors:

$$s_k^{\text{type}} \in \{\text{cooperator, reciprocator, exploiter, random}\} \quad (\text{latent behavioral type}) \quad (5)$$

$$s_k^{\text{stance}} \in \{\text{trusting, neutral, hostile}\} \quad (\text{latent stance}) \quad (6)$$

$$s^{\text{own}} \in \mathcal{A} \quad (\text{agent’s executed social action}) \quad (7)$$

The own-action state is *deterministic bookkeeping*: the agent always knows its own action, and this factor makes that knowledge available to the observation model. It is not a psychological hidden variable. In the binary task, the joint hidden state per partner spans  $4 \times 3 \times 2 = 24$  possibilities; graded regimes replace the binary action factor with the configured investment levels.

Observations are partner action and payoff:

$$o_k^{\text{act}} \in \{\text{cooperate, defect}\} \quad (\text{partner response}) \quad (8)$$

$$o_k^{\text{pay}} \in \{-1, 1, 3, 5\} \quad (\text{focal agent’s realized payoff}) \quad (9)$$

#### A.2 Partner-action likelihood

The core of the generative model is the *partner cooperation likelihood table*, which defines the probability the partner cooperates given type and stance:

Table 3: Cooperation likelihood  $P(\text{cooperate} \mid s^{\text{type}}, s^{\text{stance}})$ .

| Partner type | Trusting | Neutral | Hostile |
|--------------|----------|---------|---------|
| Cooperator   | 0.95     | 0.80    | 0.55    |
| Reciprocator | 0.90     | 0.70    | 0.30    |
| Exploiter    | 0.70     | 0.35    | 0.10    |
| Random       | 0.60     | 0.50    | 0.35    |

This table is the foundation of the likelihood modality  $A_k^{\text{act}}$ . Entries define

$$P(o_k^{\text{act}} = \text{cooperate} \mid s_k^{\text{type}}, s_k^{\text{stance}}), \quad (10)$$

with  $P(o_k^{\text{act}} = \text{defect} \mid \cdot) = 1 - P(o_k^{\text{act}} = \text{cooperate} \mid \cdot)$ . The modality is uniform over  $s^{\text{own}}$ : partner action depends only on inferred type and stance. This operationalizes what the agent considers a predictable partner—a cooperative reciprocator in a trusting stance has high predicted cooperation; an exploiter in a hostile stance has low predicted cooperation.

### A.3 Payoff likelihood

Payoff enters as a second observation modality  $A_k^{\text{pay}} = P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$  rather than as external reward. The environment payoff table is deterministic (Table 1 in the main text); in the generative model, partner action is marginalized through the cooperation likelihood:

$$P(o_k^{\text{pay}} = v \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}}) = \sum_{a^{\text{partner}} \in \{C, D\}} \mathbb{I}[\pi(s^{\text{own}}, a^{\text{partner}}) = v] P(o_k^{\text{act}} = a^{\text{partner}} \mid s_k^{\text{type}}, s_k^{\text{stance}}), \quad (11)$$

where  $\pi(\cdot, \cdot)$  maps joint actions to focal payoffs  $\{-1, 1, 3, 5\}$  and  $P(o_k^{\text{act}} \mid \cdot)$  comes from Eq. (10). This preserves the Prisoner’s Dilemma incentive structure at horizon 1 while allowing trust-building dynamics to emerge over longer horizons.

### A.4 Transition dynamics

*Type drift* ( $B_k^{\text{type}}$ ). Partner type is uncontrollable and drifts slowly:

$$P(s_k^{\text{type}'} = j \mid s_k^{\text{type}} = i) = \begin{cases} 1 - p_{\text{switch}} & j = i, \\ p_{\text{switch}} / (K_{\text{type}} - 1) & j \neq i, \end{cases} \quad (12)$$

with  $K_{\text{type}} = 4$  and default  $p_{\text{switch}} = 0.05$ . Scheduled betrayal and mixed-volatility scenarios set  $p_{\text{switch}} = 0$ .

*Stance dynamics* ( $B_k^{\text{stance}}$ ). Stance transitions are action-dependent on the focal agent’s social action  $a$ :

$$P(s_k^{\text{stance}'} \mid s_k^{\text{stance}}, a), \quad (13)$$

as specified below. Cooperation tends to build trust; defection damages trust; recovery from hostility is slower than collapse.

Table 4: Stance transition dynamics  $P(s'^{\text{stance}} | s^{\text{stance}}, a)$  conditional on focal agent action.

| <b>Agent cooperates</b> |          |         |         |
|-------------------------|----------|---------|---------|
| To \ From               | Trusting | Neutral | Hostile |
| Trusting                | 0.90     | 0.30    | 0.05    |
| Neutral                 | 0.10     | 0.60    | 0.35    |
| Hostile                 | 0.00     | 0.10    | 0.60    |
| <b>Agent defects</b>    |          |         |         |
| To \ From               | Trusting | Neutral | Hostile |
| Trusting                | 0.10     | 0.05    | 0.02    |
| Neutral                 | 0.50     | 0.35    | 0.18    |
| Hostile                 | 0.40     | 0.60    | 0.80    |

These dynamics instantiate the asymmetry of trust: cooperation gradually builds trust (neutral  $\rightarrow$  trusting in  $\sim 3$ –4 steps), while defection rapidly damages trust (trusting  $\rightarrow$  hostile in  $\sim 2$  steps); recovery from hostility is slow ( $\sim 4$ –5 steps to neutral).

*Own-action bookkeeping* ( $B^{\text{own}}$ ). The own-action factor updates deterministically from the executed social action:

$$P(s^{\text{own}'} = a' | s^{\text{own}}, a) = \mathbb{I}[a' = a]. \quad (14)$$

This factor is not a psychological hidden variable; it makes the agent’s executed action available to the payoff likelihood in standard factorized form.

### A.5 Preferences, priors, and policy priors

*Observation preferences* ( $C_k$ ). There is no direct preference over partner actions:

$$C_k^{\text{act}} = [0, 0] \quad \text{over } \{\text{cooperate, defect}\}. \quad (15)$$

Payoff preferences ascend over  $\{-1, 1, 3, 5\}$ :

$$C_k^{\text{pay}} = \log_{\text{softmax}}\left(\frac{[-1, 1, 3, 5]}{\tau}\right), \quad (16)$$

where  $\tau$  is a temperature parameter controlling preference sharpness. Instrumental motivation therefore enters through payoff observations and multi-step planning rather than direct action preferences.

*Initial-state priors* ( $D_k$ ). Type, stance, and own-action priors are

$$D_k^{\text{type}} = \left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right], \quad (17)$$

$$D_k^{\text{stance}} = [0.2, 0.6, 0.2], \quad (18)$$

$$D^{\text{own}} = [0.5, 0.5] \quad (\text{binary task; uniform over graded levels in graded regimes}). \quad (19)$$

*Policy priors ( $E_k$ ).* Policy priors are uniform over admissible action sequences at the configured planning horizon (no habitual bias toward cooperation or defection). For horizon 1 in the binary task,  $E_k = [0.5, 0.5]$  over {cooperate, defect}; longer horizons enumerate all length- $\tau$  sequences with equal prior mass.

Partner selection across  $k$  is handled outside the per-partner POMDP by evaluating policy branches for each partner with precision-adjusted logits before executing the chosen action (Appendix B).

### A.6 Partner implementation

Partner implementation follows Section 2.1. Four environment-side partners maintain latent type and stance and sample actions from the cooperation likelihood in Table 3, undergoing action-dependent stance transitions per Table 4. Partners do not perform variational inference or maintain affective precision estimates, isolating the focal agent’s precision mechanism before extending to reciprocal active-inference partners.

## B Affective Precision Update and Policy Selection

This appendix gives the  $\beta$ -update rule, policy-precision mapping, and partner-choice policy-selection procedure.

### B.1 Categorical beta support

Each partner maintains a categorical posterior  $q(\beta_k)$  over discrete levels  $\{0.5, 0.67, 1.0, 1.5, 2.0\}$ , updated each round from affective charge  $\phi_k$ . The posterior is categorical rather than continuous; this keeps the inverse- $\beta$  convention identifiable while preserving a compact implementation of the precision dynamics.

### B.2 Persistence prior

The update proceeds in three steps. First, a persistence prior smooths the previous posterior via a tridiagonal transition matrix:

$$p(\beta_k) \leftarrow T \cdot q_{\text{prev}}(\beta_k), \quad (20)$$

where  $T$  is tridiagonal with persistence parameter 0.8 and reflecting boundaries, encoding the prior expectation that precision evolves slowly.

### B.3 Charge-based pseudo-likelihood

Second, a pseudo-likelihood is constructed directly from affective charge, mapping positive charge to high precision (low  $\beta$ ) and negative charge to low precision (high  $\beta$ ):

$$\log \ell(\beta_\ell) = \phi_k \cdot \frac{1}{\beta_\ell}, \quad (21)$$

where  $\beta_\ell$  ranges over the five support points. This is a hand-designed update rule (not derived from the POMDP generative model) chosen to satisfy monotonicity: strong positive charge reinforces low  $\beta$ , strong negative charge reinforces high  $\beta$ , and the effect scales with effective precision at each level.

#### B.4 Posterior update and policy precision

Third, the posterior is normalized:

$$q(\beta_k) \propto \exp(\log \ell(\beta_\ell)) \cdot p(\beta_k). \quad (22)$$

Policy precision is then set from the posterior mean as given in Eq. 3 of the main text. The  $\beta_k$  posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent’s ability to form prior expectations over its own future affective states (discussed in Section 4).

#### B.5 Centered cross-partner policy logits

In partner-choice regimes, precision must sharpen policies within a partner branch without creating artifacts when branches are compared across partners. The centered logit procedure is given in Eq. 4 of the main text. The agent samples or selects policies from the combined set of centered logits across all partner branches. Centering preserves the mean score of each partner branch while allowing  $\gamma_k$  to sharpen or flatten the within-partner policy distribution, preventing low-precision scaling from making uniformly poor partner branches spuriously attractive in cross-partner comparison.

#### B.6 Ablation variants

The four affect-runtime variants are defined in Section 2.5. For completeness: the no-affect control disables the  $\beta$  tracker entirely and fixes  $\gamma_k = \gamma_{\text{base}}$ ; the tracked-only ablation updates  $q(\beta_k)$  normally but fixes  $\gamma_k = \gamma_{\text{base}}$ , allowing the precision estimate to be analyzed without influencing action selection; the default partner-local variant applies the full update pipeline described in B.1–B.4 and sets  $\gamma_k$  from each partner’s posterior mean; and the shared- $\beta$  ablation maintains a single pooled  $\beta$  state updated from all partner evidence while keeping per-partner POMDP beliefs intact, with all partner branches inheriting precision from the shared expected  $\beta$ .

## C Simulation Protocols and Metrics

This appendix summarizes simulation conditions, replication counts, schedules, and metrics.

### C.1 Simulation conditions

All reported experiments use one focal active-inference agent with four partners ( $K = 4$ ), per-partner generative models, and environment-side partner policies (Appendix A). Conditions differ by payoff regime, assignment mode, and scheduled partner changes.

**Binary random-assignment baseline.** In the binary trust game (Table 1 in the main text), partners are assigned randomly each round and the agent chooses cooperate or defect. Stochastic partner-type drift uses  $p_{\text{switch}} = 0.05$  unless disabled for scheduled-switch scenarios.

**Open graded decision regime.** The graded regime replaces the binary action with a discretized investment level while keeping the same latent type-stance structure. This is the primary deployment-dissociation setting (Section 3.3): affect, no-affect, and tracked-only variants are compared on policy entropy and cumulative payoff.

**Partner-choice regime.** The agent first selects a partner, then selects an action toward that partner each round. Partner selection is where engagement reorganization and social-allocation readouts are defined (Section 3.4).

**Focal-switch locality probe.** A partner-choice graded environment with type volatility disabled ( $p_{\text{switch}} = 0$ ) and a single scheduled stance switch on partner P0 tests whether local versus shared  $\beta$  tracking preserves the surprise-over-reward dissociation (Section 3.1).

**Abrupt betrayal.** A partner-choice graded environment with scheduled stance betrayal on a previously cooperative partner tests post-switch reallocation, entropy, payoff, and joint accuracy (Section 3.5).

**Precision-dynamics parameter variation.** Computational profile variants (low-gain, high-gain, and cautious-prior profiles) vary  $\beta_k$  amplitude and stability in the graded decision regime (Appendix D).

**Extended profile experiments (Exp A–D).** Parameter sweeps and factorial designs vary precision gain  $\alpha$  and prior over model fitness to define computational trust-calibration profiles in open graded, betrayal, partner-choice, forgiveness, and mixed-volatility environments (Section 3.6).

### C.2 Seed counts and episode lengths

Unless otherwise stated in the main text, episode length is 200 rounds. The abrupt-betrayal condition uses 120 rounds because post-switch readouts dominate interpretation. Profile-scale and mixed-volatility tables use 20 seeds per condition. Central behavioral comparisons use 30 seeds per condition. Exploratory diagnostic runs used three seeds per variant and were used only for implementation validation; all main-text claims are based on the confirmation-scale or profile-scale runs reported in the relevant sections.

Representative configured scales:

Table 5: Representative episode lengths and replication counts by experiment family.

| Experiment family            | Setting                                             | Rounds | Seeds |
|------------------------------|-----------------------------------------------------|--------|-------|
| H0/H2 open graded deployment | graded, random or open choice                       | 200    | 30    |
| H1 model fitness             | graded / reliability probes                         | 200    | 30    |
| H3 focal-switch locality     | partner choice, single switch                       | 100    | 30    |
| H4 partner choice            | partner choice                                      | 200    | 30    |
| H5 abrupt betrayal           | partner choice, scheduled betrayal                  | 120    | 30    |
| H6 perturbation dynamics     | graded perturbations                                | 200    | 30    |
| Exp A–D profiles             | sweeps / factorial / forgiveness / mixed volatility | 200    | 20    |

*Supplementary materials.* The supplementary repository contains the simulation environments, affect-runtime variants, experiment configuration files, analysis scripts, and figure-generation code used for the reported experiments.

### C.3 Betrayal schedule

The canonical abrupt-betrayal protocol (H5 `betrayal_choice`) uses graded pay-offs, agent-chosen partners, four partners with fixed initial types and stances, and no stochastic type switching ( $p_{\text{switch}} = 0$ ). Partner P0 begins as a cooperator in a trusting stance. At round 31, P0 undergoes a scheduled stance switch to hostile while other partners retain their initial configurations. Primary readouts are post-switch cumulative payoff, policy entropy, partner reallocation, and joint partner-type accuracy over the remaining episode.

The profile-scale betrayal environment used in Exp A and Exp B applies the same single-switch structure within 200-round episodes for recovery-time and quadrant metrics. Exp C (forgiveness) extends this schedule: rounds 1–80 cooperative baseline, rounds 81–120 betrayal on P0, rounds 121–200 reversion to cooperative/trusting on P0.

### C.4 Mixed-volatility schedule

The mixed-volatility environment (Exp D; Section 3.6) places four partners with heterogeneous volatility in the same partner-choice episode over 200 rounds:

- **P0**: stationary cooperator (no scheduled change).
- **P1**: stationary exploiter (no scheduled change).
- **P2**: abrupt shift—cooperative/trusting pre-switch, hostile/exploiter from the scheduled switch onward (round 101 in the implemented scenario).
- **P3**: gradual drift—stance and type transitions spread across mid-episode rounds (neutral stance from round 51; hostile/exploiter shift from round 151 in the implemented scenario).

Stochastic type switching is disabled ( $p_{\text{switch}} = 0$ ). Conditions compare default, low- $\alpha$ , high- $\alpha$ , and no-affect variants at 20 seeds each.



### C.5 Partner-choice allocation metrics

Partner-choice readouts quantify how precision reshapes *who* the agent engages, not only how it acts toward a fixed partner.

**Selection share.** Per-partner selection rate is the fraction of rounds on which each partner is chosen. Partner-choice reallocation is summarized qualitatively in Section 3.4.

**Selection concentration.** The Gini coefficient of the partner-selection distribution summarizes how concentrated engagement is across the social portfolio (used in profile Exp A and Exp B).

**Post-betrayal allocation.** Post-switch windows report reallocation away from the switched partner and toward alternatives, including post-betrayal P0 selection and high-investment rates in profile analyses.

**Policy entropy under choice.** Mean  $q_\pi$  entropy in partner-choice regimes captures how sharply the agent commits to partner-action policies (reported comparatively for affect versus no-affect controls).

### C.6 Entropy, payoff, and accuracy readouts

**Total payoff.** Cumulative focal-agent payoff over the episode (or over defined post-switch windows for betrayal analyses).

**Policy entropy.** Mean  $q_\pi$  entropy ('q\_pi\_entropy' in logged outputs) measures action-commitment sharpness; lower entropy indicates sharper policy selection under precision modulation.

**Joint partner-type accuracy.** Mean joint correctness of inferred partner type ('mean\_joint\_accuracy') summarizes epistemic state inference; it is reported separately from payoff because precision modulation can change deployment and accuracy independently of cumulative reward.

**$\beta_k$  range.** Per-partner or episode-aggregated range of  $\beta_k$  support summarizes precision-dynamics amplitude in profile and perturbation tables.

**Model-fitness dissociation.** Partner-local correlations between  $\beta_k$ -derived precision and action surprisal versus realized payoff test whether the confidence signal tracks predictive reliability rather than partner value (Section 3.1).

**Mixed-volatility discrimination metrics.** The discrimination index is the Pearson correlation between per-partner  $\beta_k$  trajectory and per-partner behavioral stability (higher indicates better separation of stable versus shifting partners). The P0 false-positive rate is the rolling rate at which engagement with the stationary cooperator P0 falls more than 15% below its early baseline despite P0 never changing.

**Betrayal timing metrics.** Recovery time and related latency readouts measure rounds after a switch until selection or payoff returns toward baseline thresholds (profile Exp A/B and betrayal analyses).

### C.7 Replication summary

Central behavioral comparisons in Section 3 use 30 seeds per condition. Profile-scale and mixed-volatility analyses use 20 seeds per condition. The model-fitness

dissociation (Section 3.1) uses active-encounter partial correlations to compare precision–surprise against precision–payoff associations, with shared  $\beta$  included as the relational-specificity ablation.

## D Extended Profile and Robustness Results

This appendix reports extended parameter sweeps, profile analyses, and robustness diagnostics that support the main-text results.

### D.1 Precision-dynamics parameter variation

We first vary the amplitude and stability of  $\beta_k$  dynamics to test how precision dynamics affect action commitment. These variants are computational controls rather than diagnostic categories.

Table 6: Precision-dynamics parameter variation in the graded decision regime.

| Variant        | $\beta_k$ range | Total payoff | Policy entropy |
|----------------|-----------------|--------------|----------------|
| affect         | 1.32            | 1851.3       | 8.59           |
| low-gain       | 0.24            | 1840.5       | 8.9            |
| high-gain      | 1.48            | 1859.0       | 8.1            |
| cautious-prior | 1.48            | 1877.7       | 8.4            |

Flattening the precision signal produces the clearest behavioral change. The low-gain variant compresses  $\beta_k$  range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines. Expanded-dynamics variants produce wider  $\beta_k$  movement, but payoff does not follow range monotonically. The parameter sweeps below therefore vary gain and prior structure directly.

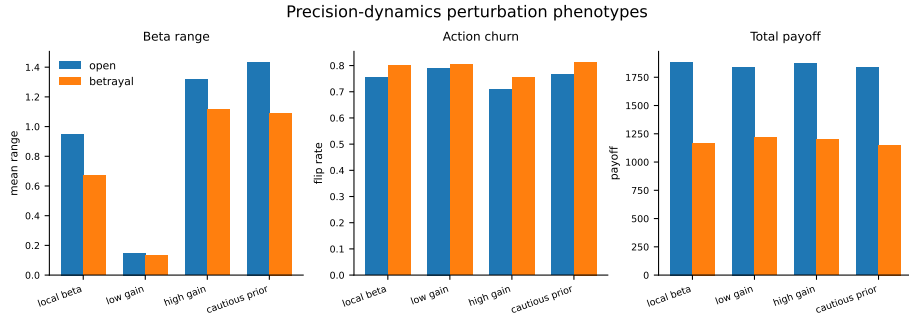


Fig. 5: Precision-dynamics parameter variation profiles in the graded decision regime. Variants compare precision-dynamics settings in the graded decision regime.

## D.2 Alpha sweep

Sweeping  $\alpha$  across  $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$  tests whether precision gain controls reactive social confidence across metrics.  $\beta_k$  movement range increases monotonically with  $\alpha$  in the betrayal regime:

Table 7:  $\beta_k$  range by precision-gain  $\alpha$  (betrayal regime, 20 seeds).

| $\alpha$ | Mean $\beta_k$ range |
|----------|----------------------|
| 0.05     | 0.147                |
| 0.1      | 0.182                |
| 0.3      | 0.326                |
| 0.5      | 0.416                |
| 1.0      | 0.580                |
| 2.0      | 0.862                |
| 4.0      | 1.051                |
| 8.0      | 1.218                |

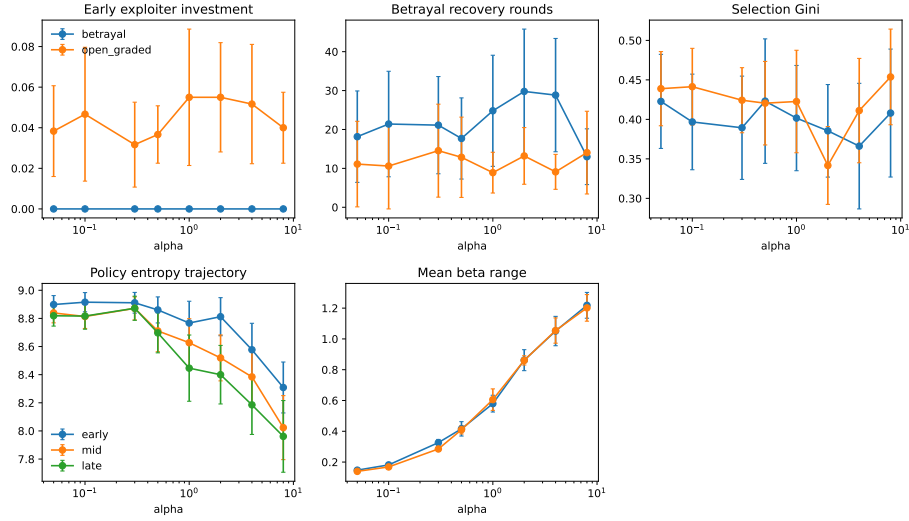


Fig. 6: Precision-gain sweep across open graded and betrayal environments (20 seeds). Higher  $\alpha$  expands  $\beta_k$  range and reshapes engagement readouts without monotonic payoff ranking.

### D.3 Profile quadrant profiles

Crossing prior belief (naive versus cautious) with  $\alpha$  (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Table 8 gives key metrics.

Table 8: Behavioral metrics by profile (betrayal environment, 20 seeds). Default agent shown for reference.

| Profile          | Payoff | Recovery | Gini  | $\beta$ range | Trust asymmetry |
|------------------|--------|----------|-------|---------------|-----------------|
| Anxious-reactive | 1932   | 26.5     | 0.338 | 1.084         | 0.98            |
| Hypervigilant    | 1951   | 20.1     | 0.423 | 0.892         | 1.65            |
| Naive-stubborn   | 1894   | 21.3     | 0.304 | 0.364         | 1.19            |
| Avoidant-rigid   | 1889   | 14.9     | 0.357 | 0.322         | 1.21            |
| Default          | 1991   | 23.0     | 0.423 | 0.893         | 1.68            |

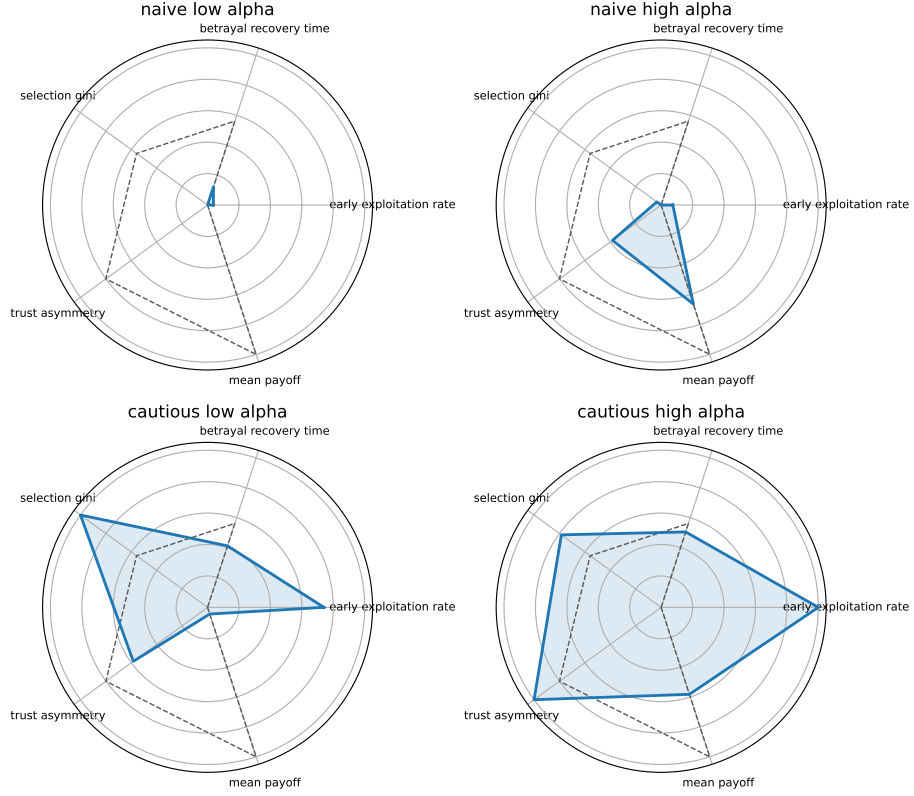


Fig. 7: Gain-prior profile quadrants from the prior-by- $\alpha$  factorial (betrayal environment, 20 seeds).

#### D.4 Forgiveness and trust repair

The forgiveness paradigm uses a 200-round partner-choice episode: cooperative baseline (rounds 1–80), betrayal on P0 (rounds 81–120), then reversion to cooperative/trusting behavior (rounds 121–200). Readouts include reengagement latency, post-return selection, payoff recovery, and  $\beta_k$  recovery trajectories for the reverted partner.

Table 9: Forgiveness and trust-repair metrics by precision profile (20 seeds). Reengagement is the post-repair P0 selection rate; payoff recovery compares late post-repair payoff with the late pre-betrayal baseline.

| Variant                 | Reengagement | Payoff recovery | Latency | $\beta_k$ range |
|-------------------------|--------------|-----------------|---------|-----------------|
| cautious-high- $\alpha$ | 0.518        | 0.996           | 5.5     | 0.971           |
| cautious-low- $\alpha$  | 0.630        | 1.014           | 1.5     | 0.287           |
| default                 | 0.475        | 1.019           | 3.0     | 1.012           |
| naive-high- $\alpha$    | 0.560        | 1.005           | 2.2     | 1.100           |
| naive-low- $\alpha$     | 0.527        | 1.004           | 4.6     | 0.325           |
| no-affect               | 0.593        | 1.033           | 1.7     | —               |

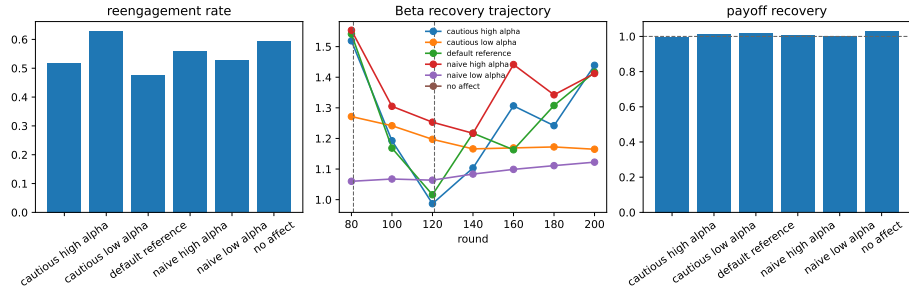


Fig. 8: Forgiveness/trust-repair diagnostic from the 20-seed profile analyses. Reengagement and payoff recovery do not collapse into a single monotonic gain effect: no-affect and cautious-low- $\alpha$  profiles reengage most often, while payoff recovery remains near baseline across profiles.

## D.5 Mixed-volatility extended diagnostics

The primary mixed-volatility table and figure remain in Section 3.6. Extended Exp D diagnostics include per-partner  $\beta_k$  trajectories, rolling P0 selection concentration, and pairwise stability–precision correlations used to compute the discrimination index and P0 false-positive rate reported in the main text.